Fluids, Pressure and Buoyancy

SYNOPSIS

Everything so far has been about solid objects — a block, a cart, a marble — things that keep their shape and that you can point at. A fluid does not keep its shape. Pour water into a different glass and it becomes a different shape; it is still the same water. So the questions we have been asking need rephrasing. “What force acts on it?” becomes awkward when the thing you are pushing spreads out under your hand.

The rephrasing is this: instead of force, ask about **force per unit area** — pressure. Instead of mass, ask about **mass per unit volume** — density. Both replace a quantity that depends on how much of the stuff you happened to scoop up with one that does not. A litre of water and a swimming pool of water have very different masses and the same density, and that is exactly why density is the useful number.

With those two, the whole of fluid statics follows from one idea: **the fluid above you has weight, and it is pressing down on you.** Go deeper and there is more of it above you, so the pressure rises. That single sentence gives you Pascal's principle, the hydraulic jack that lifts a car with one hand, and Archimedes' principle — why a steel ship floats while the same steel, melted into a lump, goes straight to the bottom.

Note 2 then sets the fluid moving.

BEFORE YOU READ ON — PREDICT

A steel ship floats. Melt that same ship down into a solid lump of steel and drop it in — it sinks. Same steel, same mass, same water. Commit to one sentence in your guided notes saying what changed, before you read on.

PERFORMANCE EXPECTATIONS

Define density and pressure and use them. Explain where the pressure in a fluid comes from, and find the pressure at a depth. Use Pascal's principle to solve a hydraulic problem. Use Archimedes' principle to decide whether something floats, and to find a density by weighing an object twice.

For PHY 151 — with calculus: Get the depth result as the solution of a differential equation rather than as a formula, and see what has to change when the density is not constant.

WATCH · THE NOTE WORKED THROUGH

Recording to come. Print the guided notes (link at the bottom of this page), watch, and fill them in as you go.

What a fluid is, and why density replaces mass

A fluid is anything that flows — anything whose molecules are free to move past one another. That takes in liquids *and* gases, which is worth saying out loud, because air behaving like a fluid is what makes half the examples in Note 2 work.

Density is mass per unit volume: $\rho = M/V$, with M in kg and V in m³, so ρ is in kg/m³. It does not depend on how much of the stuff you have — a drop of water and a lake have the same density. One conversion is worth having at your fingertips, because you will meet both forms: ρ_{water} = **1 g/cm³ = 1000 kg/m³**. The gram-per-cc form is what the lab uses; the SI form is what the equations want.

1. A 250 cm³ jug of cooking oil has a mass of 230 g. (A) Find its density in g/cm³ and in kg/m³. (B) Poured into a glass of water, does the oil end up above the water or below it? Name the comparison that decides it.
2. A student writes the density of seawater as 1.03 kg/m³. One factor is wrong. Which way is it wrong, and what is the right number?
3. A single drop of water, and a full bathtub of water. Of M, V and ρ, which are the same for both and which are not?

i) 230 g ÷ 250 cm³ = **0.92 g/cm³ = 920 kg/m³**. It sits **above** the water, because 920 < 1000 — the comparison is with the density of the fluid, never with its mass or its volume

ii) out by a factor of 1000, the wrong way. 1.03 is the g/cm³ figure, so in SI it is **1030 kg/m³** — a number in kg/m³ looks a thousand times bigger than the same density in g/cm³, and that is the usual place to go wrong

iii) only **ρ**. *M* and *V* are both enormously bigger for the bathtub; their ratio is not which is the whole reason density is the useful number

Pressure — force spread over an area

Push on a fluid and the push spreads out. What matters to the fluid is not how hard you pushed but how hard you pushed *per unit of area*: $\boldsymbol{P = F/A}$, with *F* in N and *A* in m², so *P* is in N/m², which has its own name — the **pascal**, Pa. Read the fraction and the two routes to a big pressure fall straight out of it: make *F* big, **or** make *A* small.

4. A 60 kg woman stands on one stiletto heel whose tip has an area of 1.0 cm². (A) What pressure does she exert on the floor, in Pa? (B) A 4000 kg elephant stands on four feet, each of area 0.090 m². What pressure does it exert? (C) Which is larger, and by roughly what factor?
5. Atmospheric pressure at sea level is 101 325 Pa. (A) What force does the air exert on the top of a desk measuring 1.5 m × 0.75 m? (B) The desk does not collapse. Why not?
6. A pin prick hurts and the same gentle push with a fingertip does not. The quantity doing the work is i) *F* ii) *A* iii) both equally iv) neither — it is the material of the pin.

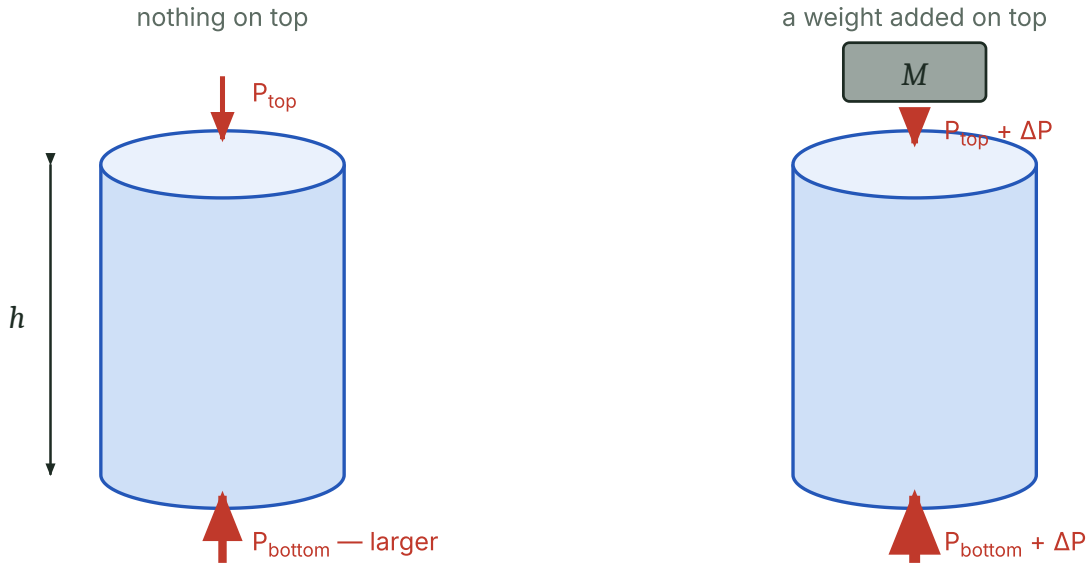
i) (A) $F = (60)(9.8) = 588$ N and $A = 1.0\ \text{cm}^2 = 1.0 \times 10^{-4}\ \text{m}^2$, so $P = \mathbf{5.9 \times 10^6\ Pa}$. (B) $F = (4000)(9.8) = 39\,200$ N over $4 \times 0.090 = 0.36$ m², so $P = \mathbf{1.1 \times 10^5\ Pa}$. (C) the **heel**, by about **54×** the area conversion is where this one goes wrong: 1 cm² is 10^{-4} m², not 10^{-2}

ii) (A) $A = 1.125$ m², so $F = (101\,325)(1.125) = \mathbf{1.14 \times 10^5\ N}$ — about 11.6 tonnes. (B) because the same air is pushing *up* underneath it and in from every side. Only a pressure **difference** pushes anything anywhere which is also why a suction cup works, and why it stops working on a rough surface

iii) **(ii).** Same *F*, a thousand times smaller *A*

Where the pressure in a fluid comes from

Nothing is pushing on a glass of water sitting on the table, and yet the pressure at the bottom of it is higher than the pressure at the top. The reason is the only one available: **the water above has weight, and it is resting on the water below.**



Two separate facts, and keeping them apart is most of this section. **Pressure depends on height** — that is the gap between the two red arrows in the left cylinder. **A change in pressure does not** — press on the top and every point in the fluid rises by the same ΔP, as on the right.

7. Take a column of fluid of height h sitting on an area A. (A) Write the extra pressure at the bottom as that column's weight divided by that area. (B) Put $m = \rho V$ into it and show that A cancels. (C) What is V/A, and what does the result become?

8. One more thing is sitting on top of the water: the atmosphere. (A) Write the **absolute** pressure at depth h. (B) How deep do you have to go in water before the absolute pressure is **two** atmospheres? Take $P_{atm} = 101\,325$ Pa and $\rho_w = 1000$ kg/m³.

9. A mercury barometer stands 760 mm tall, and mercury is 13.6 times denser than water. How tall would a *water* barometer have to be? Compare your answer with question 8 and say why the two agree.

i) (A) $P = {}^{\text{weight above}}/_{\text{area}} = mg/A$. (B) $m = \rho V$, so $P = \rho Vg/A = \rho(V/A)g$. (C) V/A is the **height** of the column — [m³] ÷ [m²] = [m] — so $\boldsymbol{P = \rho hg}$ the area cancelled: a narrow pipe and a wide lake give the same pressure at the same depth. Shape does not matter; only depth does

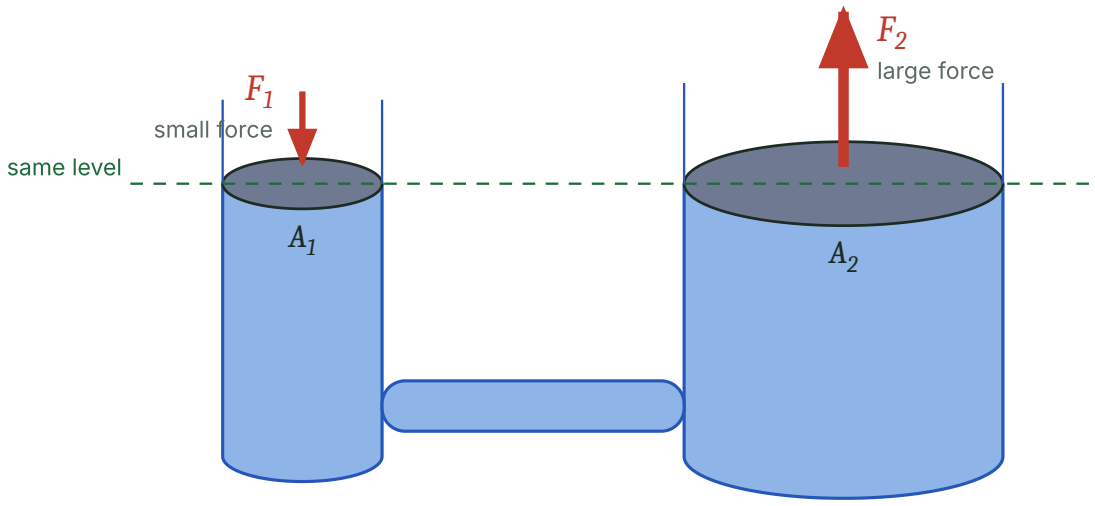
ii) (A) $\boldsymbol{P_{abs} = P_0 + \rho hg}$. (B) two atmospheres means the water has to supply one atmosphere on its own, on top of the air already there: $\rho hg = P_{atm}$, so $h = 101\,325 / (1000 \times 9.8) =$ **10.3 m** about ten metres of water is one extra atmosphere, and every further ten metres adds another — the number divers carry in their heads

iii) 13.6 × 0.760 m = **10.3 m** — the **same number**, and not by coincidence. A barometer stands at

exactly the height whose fluid column weighs one atmosphere, so the two questions are the same question asked twice say "about eleven metres" and you have lost the connection to question 8

Pascal's principle, and the hydraulic jack

The right-hand cylinder in the figure above is Pascal's principle: **a change in pressure applied to an enclosed fluid is transmitted undiminished to every part of it.** Undiminished is the word that matters — the increase does not fade with distance. The pressure itself still depends on height, but the *change* does not. That is a statement about fluids. It is also a machine.



One fluid, one pressure. Four times the area, four times the force.

The two piston faces are drawn to scale — the wide one has four times the area of the narrow one. Because both surfaces sit at the **same level**, the depth term is identical on each side and drops out, leaving $P_1 = P_2$ exactly. (The force arrows are schematic.)

10. In the figure the wide piston has four times the area of the narrow one. (A) Why is $P_1 = P_2$ here, and what would break that equality? (B) Write F_1 in terms of F_2, A_1 and A_2. (C) You push the narrow piston with 250 N. What force appears at the wide one?

11. To raise the wide piston by 2.0 cm, how far must the narrow piston travel? What does that tell you about the work done?

12. A hydraulic jack multiplies i) force ii) work iii) energy iv) all three.

i) (A) because the two surfaces sit at the **same level**, so the ρ*hg* term is identical on each side and cancels; raise one piston above the other and it no longer does. (B) $F_1/A_1 = F_2/A_2$, so $\boldsymbol{F_1}$ =

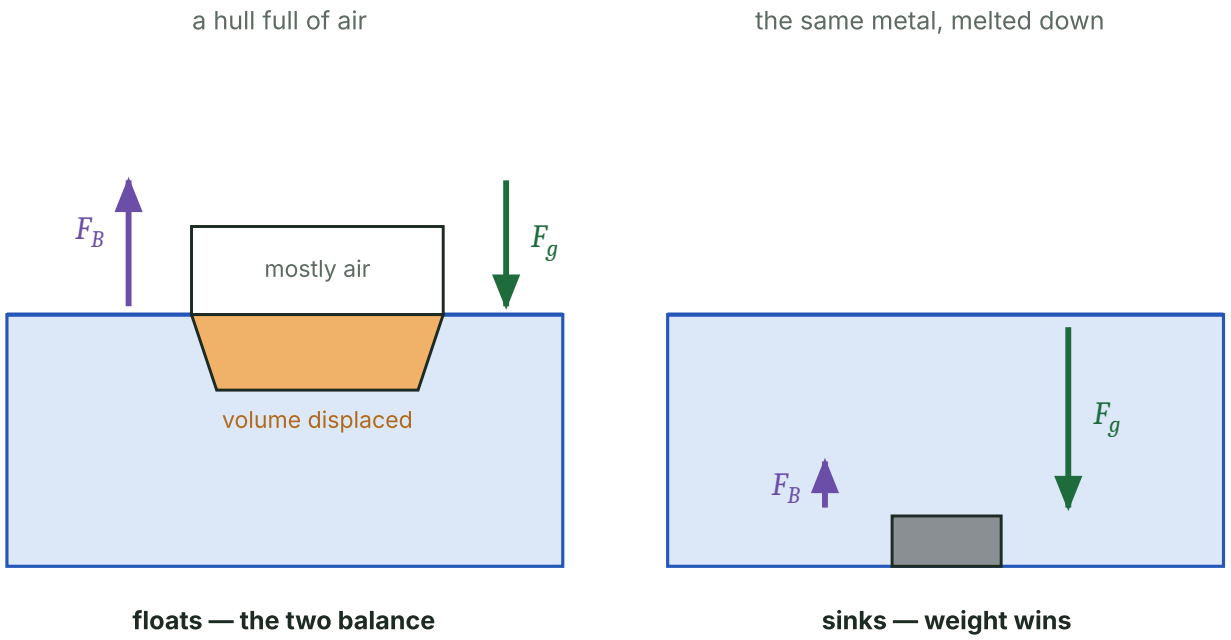
F_2A_1/A_2. (C) four times the area, four times the force: **1000 N**

ii) the fluid pushed out of one cylinder has to arrive in the other, so $A_1d_1 = A_2d_2$ and $d_1 = 4 \times 2.0 =$ **8.0 cm**. Four times the force over a quarter of the distance — the **work is the same**

iii) **(i).** Force only if it multiplied work you would have a free-energy machine; this is a car jack, a hydraulic press and the brakes on every vehicle you have been in, and none of them is one

Archimedes' principle — why the ship floats

Put something into a fluid and it has to shove some of that fluid out of the way. The fluid pushes back, and Archimedes' principle says exactly how hard: **F_B is the weight of the fluid displaced by the submerged part**, which is $F_B = \rho_w V_{dis} g$. Read the subscripts carefully. ρ_w is the density of the **fluid**, and V_{dis} is the volume of **fluid pushed aside**. Neither one is a property of the object.



Same steel, same mass, same water — and the arrows are drawn to the same scale in both panels. The hull encloses a large volume of air, so it displaces a large volume of water before it is anywhere near submerged. The lump displaces only its own small volume. **What changed is not the steel. It is the volume displaced.**

13. A floating object is in equilibrium. (A) Set $F_g = F_B$ and reduce it to $\rho_s V_s = \rho_w V_{dis}$. (B) For a *solid* object fully submerged, $V_{dis} = V_s$. What does the condition collapse to?

14. A steel ship floats; the same steel melted into a lump sinks. Steel is about 8000 kg/m³. In two sentences, using V_{dis}, say what changed.

15. A block of wood of density 600 kg/m³ floats in water. What fraction of its volume is below the surface?

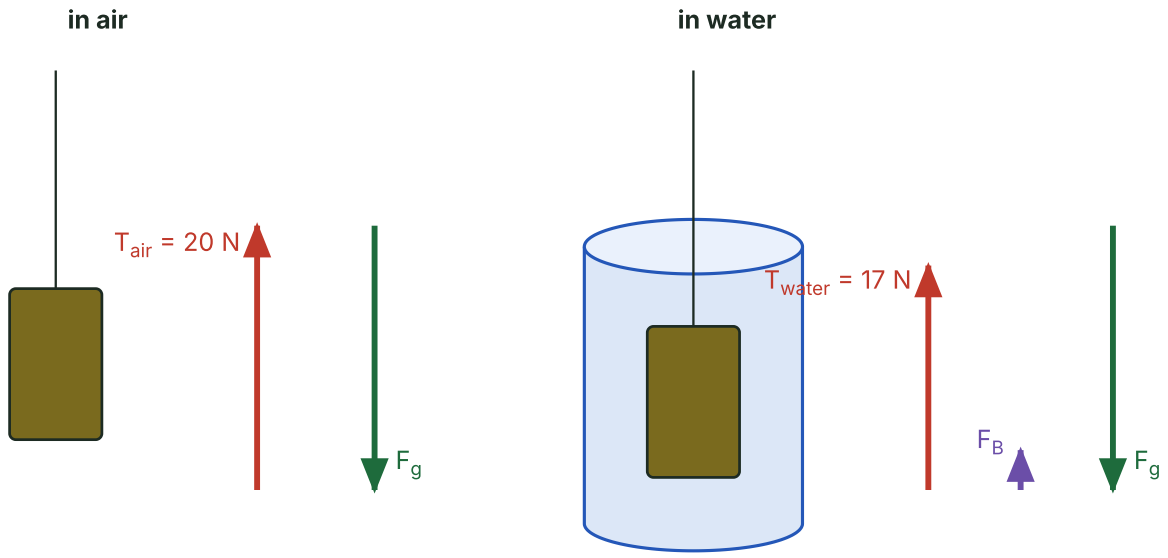
i) (A) $M_{ship}g = M_w g$, so $\rho_s V_s g = \rho_w V_{dis} g$ and $\boldsymbol{\rho_s V_s = \rho_w V_{dis}}$. (B) with $V_{dis} = V_s$ it collapses to $\boldsymbol{\rho_s < \rho_w}$ so "denser than water sinks" is not a rule to memorise any more; it is a consequence

ii) not the steel — the **volume displaced**. A hull is mostly air, so it pushes aside a large volume of water, and therefore a large weight of water, long before it is submerged; melted into a lump the same steel can only ever displace its own small volume, which weighs about an eighth of what the steel does 8000 against 1000 kg/m³

iii) $\rho_s V_s = \rho_w V_{dis}$ gives $V_{dis}/V_s = 600/1000 = \mathbf{0.60}$ — sixty per cent of it is under

Weighing something twice to find its density

Here is the part of Archimedes that earns its keep in the laboratory, and it is what you will do in the Buoyancy Lab. Hang a lump of metal from a spring scale and read the tension. Then lower it into water, fully submerged, and read the tension again. It weighs less — not because its weight changed, but because the water is now helping to hold it up. What the scale reports is the **apparent weight**.



The weight is the same in both. The scale reads less because the water carries part of it.

The two green arrows are identical lengths, because the weight has not changed. The difference between the two scale readings *is* the buoyant force.

16. A lump of metal hangs from a spring scale reading 20 N in air and 17 N fully submerged in water. (A) Draw the free-body diagram for the submerged case and write the equilibrium equation. (B)

Find F_B.

17. Now get a density out of it. The metal is *fully* submerged, so the water it displaces has exactly the metal's own volume V. (A) Write F_B as $\rho_w Vg$ and T_{air} as $\rho_m Vg$, and eliminate V and g. (B) With $\rho_w = 1000$ kg/m³, find ρ_m.

18. What is the hard measurement this method avoids, and why does that matter in the Buoyancy Lab?

i) (A) three forces now instead of two — T_{water} up, F_B up, F_g down — and T_{air} *is* the weight, so $T_{water} + F_B = T_{air}$. (B) $\boldsymbol{F_B = 20 - 17 = 3}$ **N**

ii) (A) $\rho_w Vg = T_{air}(1 - T_w/T_a)$ and $T_{air} = \rho_m Vg$, so $\boldsymbol{\rho_w = \rho_m(1 - T_w/T_a)}$ — V and g both cancel. (B) $1 - 17/20 = 0.15$, so $\rho_m = 1000/0.15 = \boldsymbol{6.7 \times 10^3}$ **kg/m³**

iii) the **volume**. Volume is the hard thing to measure on an irregular object, and this method never needs it — two scale readings and one known density give you the other one run it the other way, known water and unknown sample, and you have measured a density without ever measuring a volume

FOR PHY 151 — WITH CALCULUS

The depth result again, as a differential equation. Question 7 built $P = \rho hg$ by weighing a column. Here is the same result got the other way. Take a slab of fluid of thickness dy at height y. The pressure difference across it supports the slab's own weight, which gives $dP/dy = -\rho g$ — minus, because pressure falls as you go up. Written this way it is obvious what the constant-density assumption was doing.

19. Integrate $dP/dy = -\rho g$ from the surface down to depth d, with ρ constant, and confirm that you recover $P_0 + \rho gd$.

20. If instead $\rho = \rho_0 e^{-y/H}$, carry out the integral and say what $P(y)$ looks like. Which assumption of question 10 has just been dropped?

i) $\int dP = -\rho g \int dy$ gives $P = P_0 - \rho gy$; at $y = -d$ that is $\boldsymbol{P_0 + \rho gd}$ the same formula as above, with the sign absorbed into "depth" instead of "height"

ii) ρ constant is what let ρ come outside the integral. With $\rho = \rho_0 e^{-y/H}$ the integral no longer collapses to a product and gives $\boldsymbol{P(y) = P_0 e^{-y/H}}$ — an atmosphere whose pressure falls off

roughly **exponentially** rather than linearly which is why “ten metres is one atmosphere” works in water and says nothing useful about air

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Where this sits. Everything on this page is fluid standing still. Note 2 sets it moving, and two new rules take over: the fluid has to keep going somewhere (continuity) and its energy has to balance (Bernoulli). The method of this last section is what you will carry into the **Buoyancy Lab**.

BACK TO YOUR PREDICTION

What changed is the **volume of water displaced**, not the steel. A hull is mostly air, so it pushes aside a large volume of water — and therefore a large weight of water — long before it is submerged. Melted into a lump, the same steel can only ever displace its own small volume, which weighs far less than the steel does. If you said “the shape changed”, you were right, but the reason shape matters is that shape sets V_{dis}.

Check yourself

1. A block measuring 2.0 cm × 3.0 cm × 5.0 cm has a mass of 81 g. What is its density, in kg/m³?

_____ kg/m³

2. You press down on the piston of a sealed cylinder of water. The pressure increase at a point near the bottom, compared with the increase near the top, is

- ○ larger, because pressure grows with depth
- ○ the same — a change in pressure is transmitted undiminished
- ○ smaller, because the water above absorbs some of it
- ○ impossible to say without the cylinder's area

3. The buoyant force on a fully submerged object depends on

- ○ the mass of the object
- ○ the density of the object

- ○ the density of the fluid and the volume displaced
- ○ the depth it is held at

4. A sample hangs from a scale reading 12.0 N in air and 9.0 N fully submerged in water. The buoyant force on it is

- ○ 21.0 N
- ○ 3.0 N
- ○ 9.0 N
- ○ 12.0 N

Your notes

BEFORE THE VIDEO

Guided notes — blank (PDF)

print or open on a tablet; record your prediction, then draw the figures and work the problems as the video goes through them.

AFTER THE MODULE

Completed notes (PDF)

released when Module 6 closes.

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